Mathematics Revision Test 15

1. $7 \times y = 98$
   What is y? _____

2. 1 cm =_____mm

3. Add 3 tens, 7 fives and 6 twos _____

4. Write 5300g to the nearest kg _____

5. Add twenty-five to eighty _____

6. $9 \times 7 =$ _____

7. $\frac{63}{9}$ = _____

8. 5 days = _____ hours

9. Multiply £1.99 by 3 _____

10. What is half of nine thousand?
    In figures:_____

1. 1 Litre – 450 ml = _____

12. 210×10 = _____

13. 90×9 = _____

14. $(\frac{1}{6}$ of 12) + $(\frac{1}{3}$ of 12)
    = _____

15. $\frac{1}{10}$ of 1km = _____

16. Multiply 2.5 by 6 _____

17. $3\frac{3}{4}$ km = _____ m

18. Reduce 125 by 20 _____

19. 2.9×10 = _____

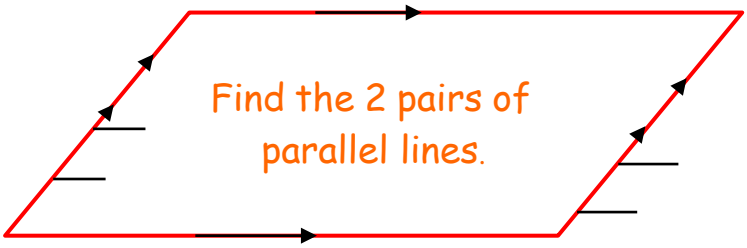
20. 8×9 = _____

Question: Has a parallelogram a line of symmetry? ◄
Cut out a parallelogram and try to fold it in half to find out.
Answer: **NO!**

Note
This question appears quite often on an exam paper.

Parallelogram Facts

A Parallelogram is a quadrilateral with both pairs of opposite sides equal and parallel.
We can draw one using 4 equilateral triangles.



Obtuse Angle = greater than a right angle.
Acute Angle = smaller than a right angle.

- In the Parallelogram above find the two obtuse angles and colour them in blue.
- Find the two acute angles and colour them in red.

In the space below draw your own Parallelogram from 4 equilateral triangles. It should just fit in the space. They should be connected.
Show all the details carefully. Be neat.

Can you spell p a r a l l e l ?
Tick the Parallelogram if you can.

Revision of volume

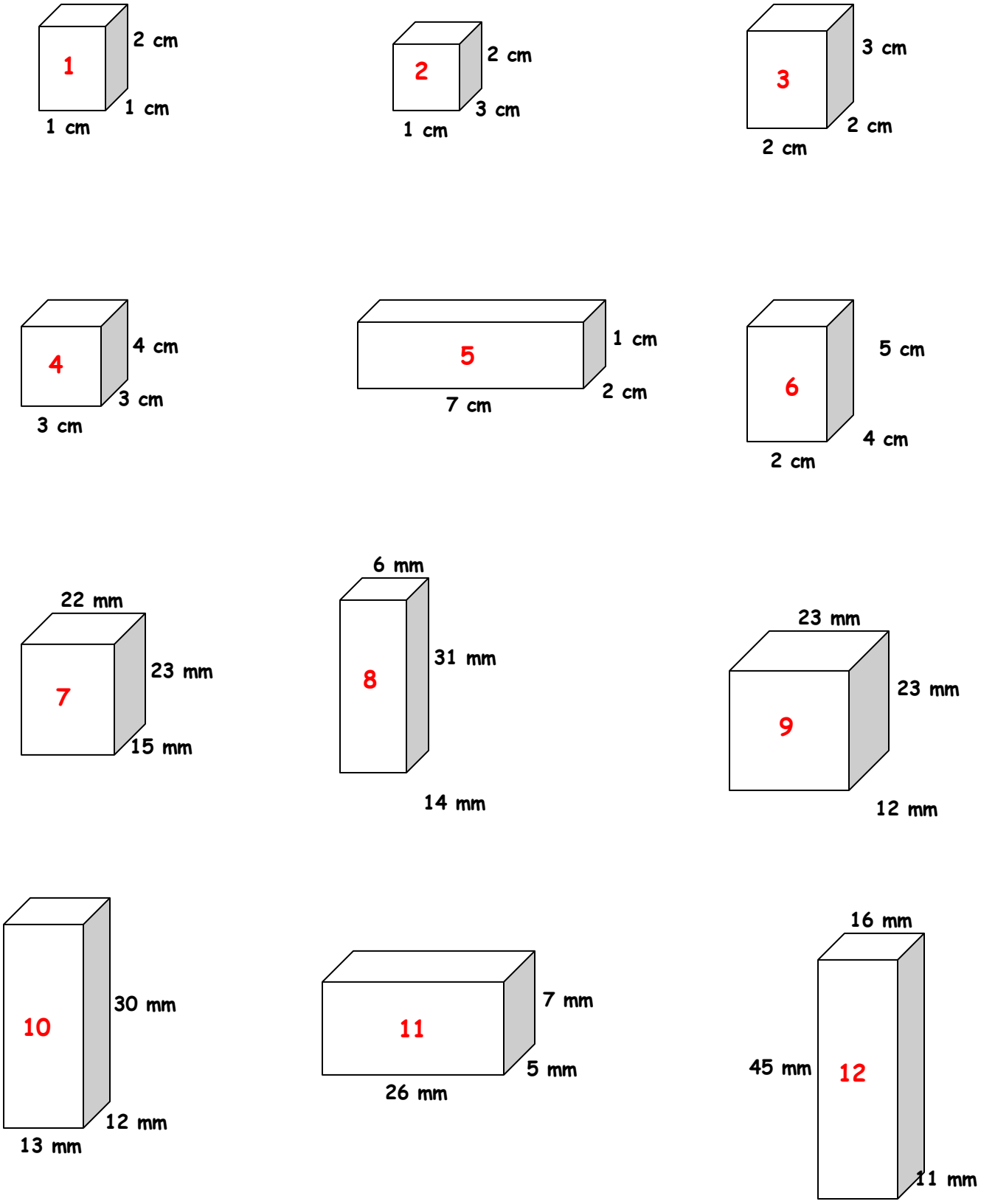
Remember: Volume = length x breadth x height

For example: 2cm x 2cm x 2cm = $8cm^3$ (8 cubic centimetres)

Now work out the volumes below from the measurements given:

(Make sure you know your long multiplication before you start or you will get in a pickle!)

Note: These cubes are not drawn to scale



Core Topic Timed Test 1

You have 30 minutes to complete this test.

1. Look at these numbers. The dotted line joins pairs of numbers which have a difference of 3. Continue the dotted line to join another pair of numbers:

10	15	12	11
7	6	9	5
0	3	19	13

Write +, −, x or ÷ in each square to make the calculation correct:

2. 5 ☐ 4 = 20
3. 9 ☐ 6 = 54
4. 100 ☐ 25 = 4
5. 15 ☐ 15 = 30
6. 40 ☐ 9 = 31
7. 36 ☐ 12 = 3
8. 55 ☐ 11 = 5
9. 34 ☐ 66 = 100
10. 53 ☐ 20 = 33

11. Write each of the numbers in the correct place in sequence:

212 344 434 221 121 209 252 343

12. to 14. Draw a ring round each number which can be divided exactly by 10:

160 34 26 90 13 100 99

15. Find 2 numbers which total 93. Draw a ring round each one.

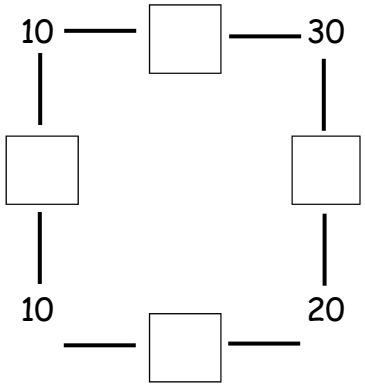
18 22 42 71 26 31 39

Core Topic Timed Test 1

16. to 19.

This is a square with some numbers missing. The numbers along each edge must add up to 70. Put each of these numbers in one of the small squares to make the totals correct:

50 20 30 40

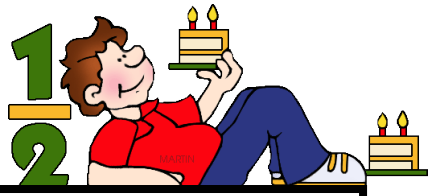


20. I think of a number. If I take away 14 it leaves 72. What is the number?
Answer:

Complete each of these number lines by following the instruction.

21.	4 less than	24	20	16	12		
22.	7 more than	14	21	28	35		
23.	Multiply by 2	3	6	12	24		
24.	Divide by 3		81		9	3	1
25.	Add 20	0	20	40	60		

Maths Reasoning



These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

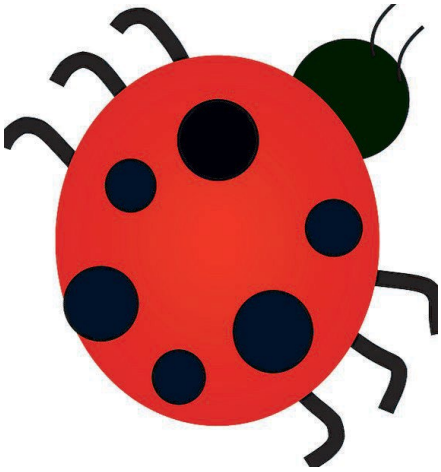
1) Ladybirds have 6 spots.

Butterflies have 7 spots.

Altogether some ladybirds and butterflies have 60 spots.

How many ladybirds are there?

How many butterflies are there?



2) What if ladybirds had 5 spots and butterflies had 8 spots and there were 160 spots altogether?

How many solutions are there?

ANSWERS - COMPLETE MATHS – PART 16

Maths
Revision sheet 15

1. 14
2. 10
3. 77
1. 5 kg
2. 105
3. 63
4. 7
5. 120
6. £5.97
7. 4,500
8. 550 ml

9. 2100
10. 810
11. 6
12. 100 m
13. 15 or 15.0
14. 3750 m
15. 105
16. 29
17. 72

Volume

1. 2 cm^3
2. 6 cm^3

3. 12 cm^3
4. 36 cm^3
5. 14 cm^3
6. 40 cm^3
7. 7590 mm^3
8. 2604 mm^3
9. 6348 mm^3
10. 4680 mm^3
11. 910 mm^3
12. 7920 mm^3

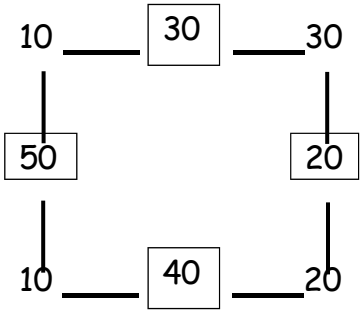
Core Topic Timed Test 1

1. 9 - 12 – 15
1. x
2. x
3. ÷
5. +
6. -
7. ÷
8. ÷
9. +
10. -
11. 121 209 212 221 252 343 344 434

12 To 14. 160, 90. 100,

15. 22, 71

16 To 19.



20. 86
21. 8, 4
22. 42,49
23. 48, 96

24. 243, 27
25. 80,100

Maths Reasoning

1) Write down the number of spots of each insect up to 60 spots.

The exact total can be determined by matching up the number of spots of each insect to exactly 60.

Number of each insect	Number of ladybird spots	Number of butterfly spots
1	6	7
2	12	14
3	**18**	21
4	24	28
5	30	35
6	36	**42**
7	42	49
8	48	56
9	54	-
10	60	-

From the table we can see that 3 ladybirds and 6 butterflies will make a total of 60 spots , so this is the correct answer.

10 ladybirds and 0 butterflies will also make a total of 60 spots.

2) Write down all the multiples of 8 from 0 to 160.

Where a multiple of 8 ends in a 5 or a 0, a combination is possible.

Number of each insect	Number of spots on butterflies
0	0
1	8
2	16
3	24
4	32
5	40
6	48
7	56
8	64
9	72
10	80
11	88
12	96
13	104
14	112
15	120
16	128
17	136
18	144
19	152
20	160

Using the table above we can see the following combinations are possible:

- 0 butterflies, 32 ladybirds
- 5 butterflies, 24 ladybirds
- 10 butterflies, 16 ladybirds
- 15 butterflies, 8 ladybirds
- 20 butterflies, 0 ladybirds

Therefore there are 5 solutions in total, but remember the original question says that there are <u>some</u> of each insect. So the actual answer is 3 solutions.

Always read questions carefully.